6 On-chain and off-chain data

This chapter covers

- Exploring different kinds of on-chain data: blocks, transactions, receipts, and state
- Defining, emitting, and logging events
- Accessing event logs from transaction receipts to support Dapp operations
- Designing and developing Dapps with on-chain and off-chain data
- Demonstrating on-chain and off-chain data using ASK and blind auction Dapps

This is it. This is what distinguishes blockchain application development from that of non-blockchain applications: *on-chain data*. Do you wonder where the data associated with a Dapp is stored? Some are stored on the blockchain infrastructure (on-chain), and others in traditional databases and files (off-chain). In this chapter, you'll learn first about the concept of on-chain data introduced by the inclusion of blockchain features in an application. Then you'll learn to design and develop Dapps that deal with a combination of on-chain and off-chain data.

So what exactly are these two types of data in the context of blockchain programming? In general, any data stored on the blockchain is called on-chain data, and anything else is off-chain.

Let's analyze this concept further. In a traditional system, the results of function executions in an application are persisted in a